

Preliminary approach to assess the seismic hazard on the Moon

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Abstract—The interest and desire of humanity to build extraterrestrial habitats brings with it great scientific and engineering challenges. To guarantee the safety and resilience of these habitats, it is necessary to characterize the hazards to which they will be exposed. Based on seismic information collected by The Apollo space program between 1969 and 1977 was developed a database tool with the purpose of allowing any researcher to manipulate and process the seismic signals of the Moonquakes. On the other hand, this paper presents an overview to assess the seismic hazard at a specific site on the Moon, using a probabilistic seismic hazard model (PSHA). To perform this analysis, only shallow Moonquakes are considered, since these are the biggest in magnitude and most energetic type of Moonquake recorded. Taking into account there have been registered only 28 shallow Moonquakes, which do not provide complete information, a Kanai-Tajimi filter is used to generate probable artificial accelerograms that adjust to the observed events

Index Terms—Resilient habitats, seismic hazard, Kanai-Tajimi filter, synthetic accelerograms.

I. INTRODUCTION

The arrival of humanity to the Moon in 1969 showed a real possibility of reaching and inhabiting satellites and planets. The design of a permanent and sustainable extraterrestrial human settlements represents a tremendous scientific and technological challenge for humanity. Habitats should be resilient under different hazards such as radiation, temperature fluctuation, meteorite impacts, and seismic activity. The later is of particular interest due to its unique characteristics. However, there is still a lack of sufficient data available to appropriately

assess the seismic activity and its effects on the design of a permanent habitat.

The lunar seismic network, consisting of four stations, was established during the Apollo program missions from 1969 to 1972, operating until September 30, 1977. This network revealed that the Moon is seismically active. The seismic waves on the Moon are less attenuated than on Earth on account of the fractured lunar regolith and the absence of water [1]. A seismic event or moonquake can last more than 10 minutes, with a slight decrease that can take hours. The seismic stations recorded more than 12,500 moonquakes, some with magnitudes above 5, highlighting the need for characterization of the seismic hazard. Therefore, a realistic analysis of the seismic activity, including seismic hazard assessment, lunar geology, and synthetic moonquakes, should be performed to accurately incorporate them in a resilient design of a future settlement.

The contribution of this paper is to serve as basis for space agencies and private companies with synthetic Moonquakes. That provides accelerograms, response and design spectra proposed in the designs of the different structures on the Moon. The organization of this paper is as follows. In Section 2, the seismic activity on the Moon is studied, considering database processing it is observed that there is no sufficient information, therefore, in Section 3, the generation of synthetic Moonquake accelerograms is being developed. Otherwise based on the seismic information collected by The Apollo space program and synthetic Moonquake accelerograms characterization of the seismic hazard is being made. Finally, the main findings

are discussed and potential applications for this model are considered in the Conclusions.

II. DATABASE DEVELOPING AND PROCESSING

A. Database

Missions 11, 12, 14, 15, and 16 of the Apollo program carried out the Apollo Passive Seismic Experiments (PSE), between 1969 and 1973 [2]. These experiments consisted of reading ground motion data through a package of seismometers and electronic equipment. Each package consisted of two seismometers, a long period one measuring a vertical component (LPZ), and two horizontal components (LPX, LPY) of the movement of the Lunar surface. The other seismometer consisted of a short period sensor that measured a vertical component (SP). [3]. In these eight years were registered 12549 events, classified in different categories, meteoroid impact, shallow Moonquakes, deep Moonquakes and undefined Moonquakes [2], [4], [5].

This investigation carries out a database with events, divided in year, day, and type. With the purpose of presenting to all scientific community and making a simple information access, giving the possibility to analyze seismic signals.

B. Processing

Each seismic signal requires two types of processing. Due to different aspects, the signals have some irregular data. In that way, when a signal is plotted, occasional spikes appear. These spikes do not represent the real information of the Moonquakes. Owing to this, an algorithm was developed. This algorithm selects the picks and gives the option to delete them, with the purpose of having the information of the signal as accurately as possible. Otherwise, there is a processing that can be made by the user. There is the option to resample the signal, with the idea to compare the Moonquake's information with other quakes. Besides, it can be obtained Moonquake response spectra. As is made in earthquakes, in Moonquakes the signal is processed applying low-and high- pass filters.

C. Comparison with earthquakes

In order to assess the lunar seismic hazard is necessary an adequate study of the seismic ground motion characteristics generated in the lunar environment. However, in the analysis of the lunar seismicity, the problem is related to the time lapse that exists in the seismic events, since is very short in comparison with the time of the earthquakes. For this reason, the understanding of the nature of the lunar seismic phenomena through the seismograms obtained by the Apollo program is very limited. Therefore, to improve its understanding we propose a comparison of the temporal, energetic and spectral characteristics of the Moonquakes with the terrestrial seismic events of tectonic and volcanic origin Fig.1 with the purpose of recognizing similarities and differences. In this way, the mechanisms of generation and propagation of lunar seismic events can be better understood. To extract and analyze the Moonquakes information an application in the MATLAB Fig.2 has been developed. Some

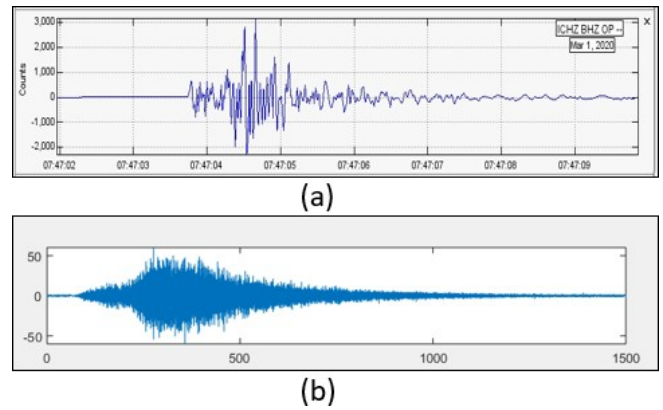


Fig. 1: Comparison between a (a) Volcanic Earthquake and (b) a Shallow Moonquake

seismic parameters such as date, localization of hypocenter, seismogram, duration, magnitude, frequencies response and identification of P and S phases can be calculated.

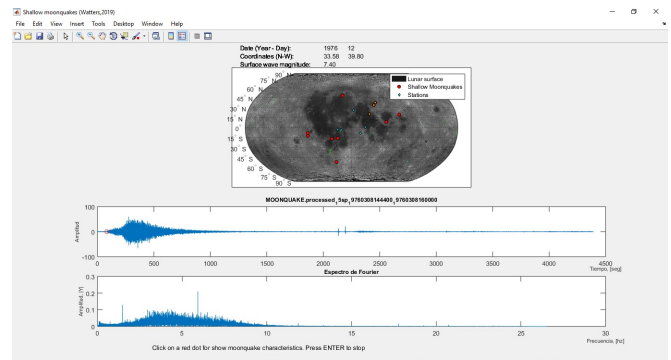


Fig. 2: App in MATLAB to determinate seismic parameters of Moonquakes

III. SEISMIC HAZARD

To guarantee the safety of a structure on a site, it is necessary to determine the maximum intensity of ground motions due to seismic events that can occur in a specific period. The probability of exceeding that level of movement is what is known as seismic hazard [6]. The hazard is quantified in terms of seismic exceedance rates or their inverses, return periods, which are related to the expected useful life of the structures at the study site.

To estimate the hazard on the Moon, it is necessary to use the instrumental information available on the seismic activity of the study area and implement one of the well-known calculation methods. In this work, the Probabilistic Seismic Hazard Assessment (PSHA) methodology is implemented due to the conceptual advantages with respect to other methodologies, since it involves uncertainties in the calculations, which is convenient in this case.

A. Calculation methodology

To assess the seismic hazard on a specific region on the Moon, the traditional PSHA method follows the next steps:

- 1) Identify and Characterize the seismic sources of Shallow Moonquakes, capable of producing dangerous ground motions.
- 2) Definition of the temporal recurrence model by which the relationship between the frequency of earthquakes and their magnitude is established.
- 3) Determination of attenuation laws associated with each source. This model relates the movement of the terrain at the site with the parameter that establishes the size of the Moonquake at the source, the source-location distance r , and a measure of the dispersion of the model.
- 4) Resolution of the probability equation to compute the seismic hazard. This equation makes use of the total probability theorem, in which uncertainties of the size of the earthquake, the location and the intensity of the ground motions are combined.

The presented methodology can be extrapolated to obtain design parameters related to seismic activity, to different site conditions and type of structures such as underground habitats in lava tubes or surface structures made of regolith and other materials [7]–[9].

B. Lunar geology and morphology

Shallow Moonquakes are the objective in this work to characterize the seismic hazard. These are the rarest in occurrence, only four detected annually by the seismic network [4], [10]. However, they represent the sources of the highest energy released. To characterize the seismic sources of shallow Moonquakes, it is important to consider the following lunar elements:

Impact Craters

They have a ring form, a central base, and peak, which a diameter that can be varied from centimeters to 250 km. They are formed principally by the impact of moving particles. Their morphology complexity is proportionate to its size, and it may affect the way that seismic waves propagate through the lunar surface. Therefore, it is important to characterize them.

Lobate scarps

These are considered the surface expressions of thrust faults. They probably formed due to contraction processes of the Moon, caused by internal cooling. Recent evidence suggests that Shallow Moonquakes could be caused by lobate scarps activity [1]. Therefore, in this study some representative faults of this type are modeled to obtain seismic parameters that will be included later in the attenuation models considered.

Study Area

The coordinates of the study zone are 21.3°N, 30°E y 19.4°N, 31.7°E. In this area the Lee-Lincoln fault scarp is located. This fault scarp extends through the North Massif with 55 km length. Based on the photographs provided by

the Lunar Reconnaissance Orbiter (LROC), it was possible to observe geological evidence that indicates that this fault and other similar ones are still active [1].

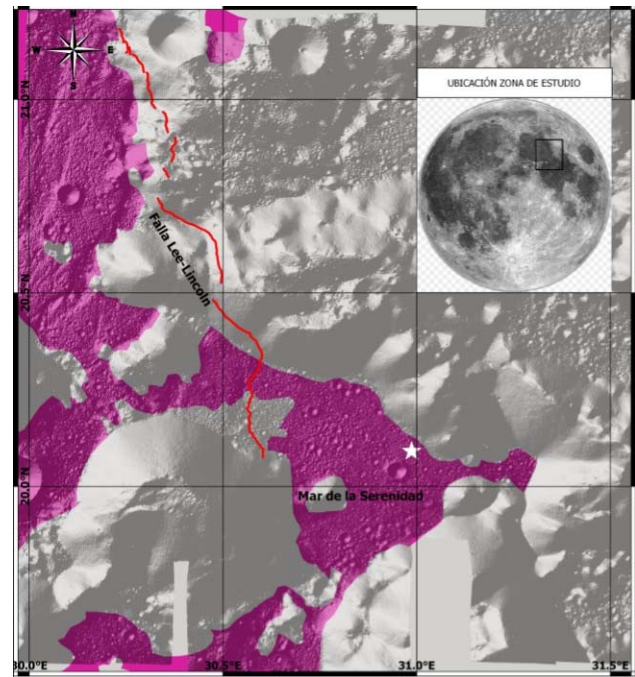


Fig. 3: Location of the Study area. Background is a DEM from LROC (2018)

C. Synthetic Moonquake accelerograms

The estimation of the seismic hazard at a site is highly dependent on the availability of the seismic data. However, in cases where there are not enough seismic information data, procedural alternatives should be taken. For this paper, synthetic accelerograms are developed based on the recorded signals and characteristics of the site. Synthetic Moonquakes are obtained using a modified Kanai-Tajimi filter, by carrying a white noise signal and passing it by a high-pass filter and a lowpass filter [11]. The obtained signal is stationary. With the purpose of getting a non-stationary signal, it is multiplied by an intensity function in the time domain. This provides a nonstationary signal like a real Moonquake.

To get better results an optimization algorithm is used. This algorithm is based on particle swarm optimization (PSO) [12]. In the beginning, the population is disorganized. Each individual has a function, the nearest particles tend to be grouped, while each individual moves randomly. With this algorithm is expected that particles move to the best solution.

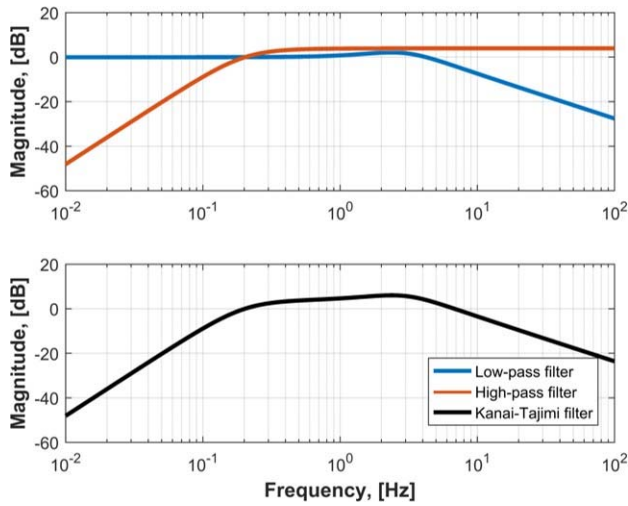


Fig. 4: Kanai-Tajimi filter Representation

IV. RESULTS

A. Database

The Moonquake's data used was founded on the Darts website in the next link: <http://darts.isas.jaxa.jp/planet/seismology/apollo/app/> (access June 30, 2020). Once this information is downloaded an excel sheet appears in one column. However, in this format it is difficult to work with it. In this way, all Moonquake events are presented in an application where can be seen in an easier way Fig.5. With this process, some spikes that do not represent the signal may be seen Fig.6.

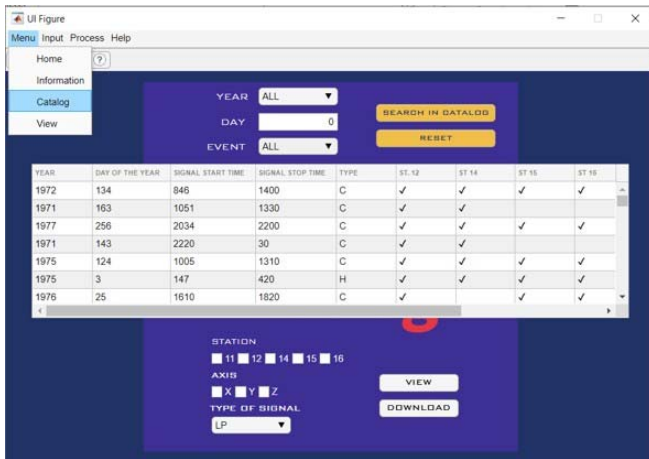


Fig. 5: Data base application

B. Processing results

In the signal processing of a Moonquake was found that the sampling period was not constant, in this way it is decided to use a constant period. Additionally, an algorithm to delete the signal peaks was applied. With the idea to do not lose information it was used an average with the data before and after the spike and change it for data that represent the signal. Fig.7 shows the processed signal.

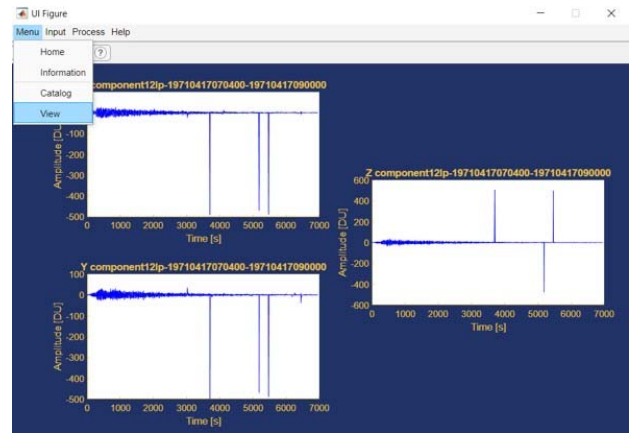


Fig. 6: Lp Moonquake signal 1971 X,Y and Z components

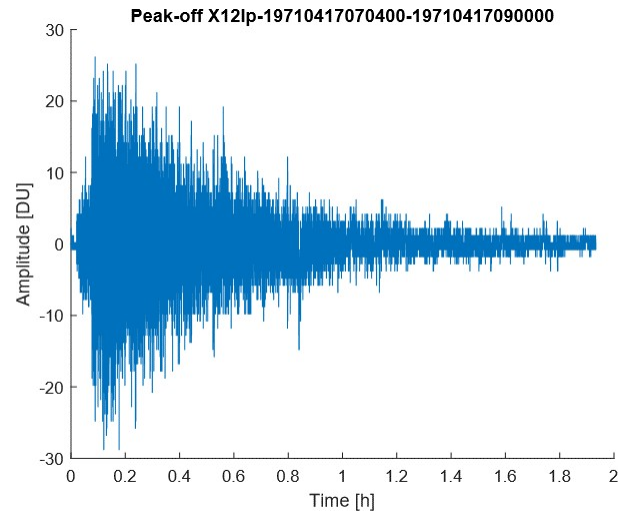


Fig. 7: Proceed Lp Moonquake signal 1971 X

C. PSO

The optimization algorithm was applied. With the purpose of having efficient results. This method develops a faster process and the obtained data were similar in both, time, and frequency domain, to real Moonquakes data.

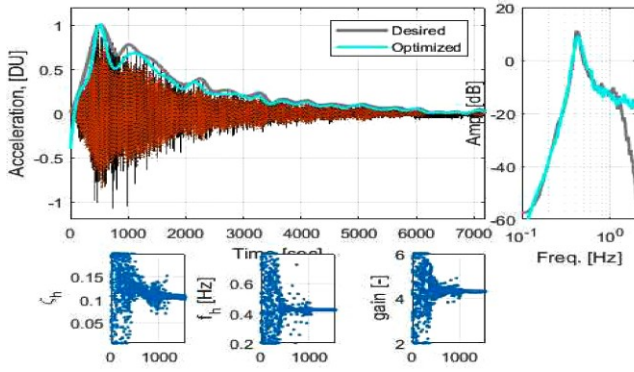


Fig. 8: Aspect of the optimized signal

D. PSHA

1) *Recurrence Model*: The magnitude recurrence is calculated for the recorded events. For this, the relation of the accumulated frequency of the earthquakes and the magnitude is used. It performs a linear regression on the data to fit the Gutenberg-Richter (GR) distribution model (1).

$$\text{Log}(N) = a - bm \quad (1)$$

where N is the cumulative number of earthquakes with a body wave magnitude m equal to or greater than the defined magnitude, and where a and b are constants that describe the decay of the frequency of occurrence of events with increasing magnitude and are found by linear least-square regression.

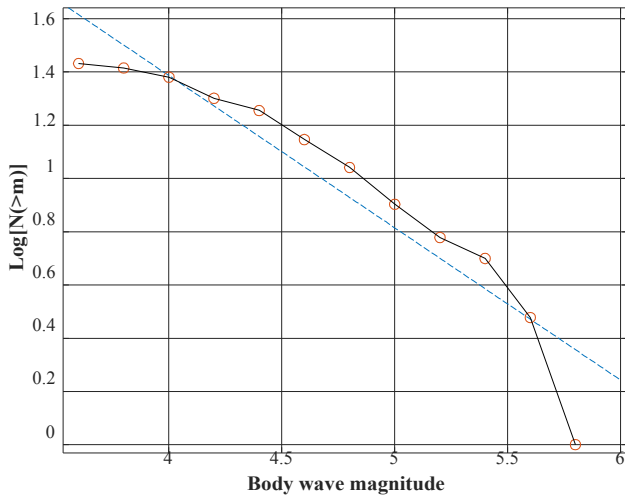


Fig. 9: Logarithm of the accumulated number of Moonquakes per year vs the body wave magnitude (Gutenberg-Richter relationship). The solid black line indicates the accumulated frequencies of the experimental data and the dotted line is the adjusted GR model, with $a = 3.68$ and $b = 0.57$.

2) *Attenuation*: Once the activity rate of the seismic sources has been determined, it is necessary to evaluate the effects they produce at the site of interest. The process focuses on

estimating the peak ground acceleration (PGA) and pseudo acceleration (SA) at different spectral periods. A few ground motions models or attenuation laws are tested and selected, according to the dispersion of the synthetic data generated.

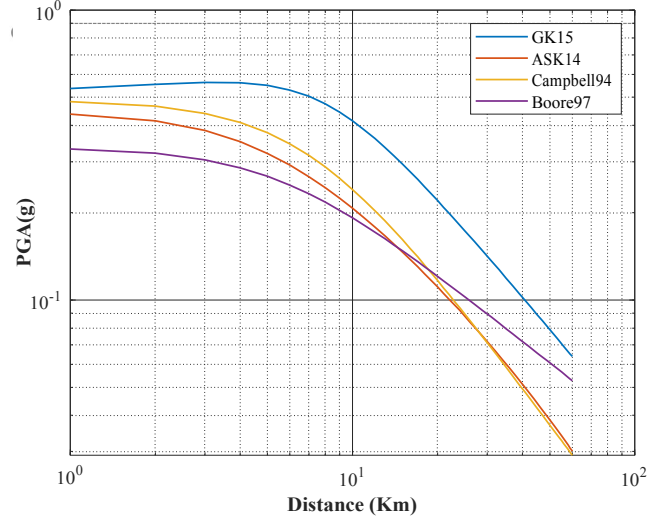


Fig. 10: Different 5% damped Ground Motion Models tested for PGA

3) *Hazard calculation*: Once the seismicity of the sources and the attenuation parameters of the generated waves are known, the seismic hazard can be calculated considering the sum of the effects of the seismic sources and the distance between each source and the site [13], as shown in the Equation 2, where $\lambda_{y\Box}$ is the exceedance rate for a given motion parameter $y\Box$, in this case, the parameter is PGA and SA at different spectral periods.

$$\lambda_{y\Box} = \sum_{i=1}^{N_s} \sum_{j=1}^{N_m} \sum_{k=1}^{N_r} v_i P[Y > y * |m_j, rk] P[M = m_j] P[R = rk] \quad (2)$$

By taking the acceleration at different periods but for the same exceedance rate, it is possible to build a uniform hazard spectrum (UHS), which could then be used in a response spectrum analysis of future structures on the Moon

V. CONCLUSIONS

The seismic data recorded by the seismometers of the missions of the Apollo program suggests that Moonquakes constitutes a significant hazard to future lunar settlements. Nevertheless, the seismic information available is not sufficient to use in structural design. The use of synthetic ground motions signals are a reliable option to solve this problem since it allows to obtain signals for different site conditions and locations. The implementation of the Kanai-Tajimi model with an algorithm based on particle swarm optimization used in this work, has shown good results to replicate Moonquakes and the shape of the power spectrum within a 0.1 Hz to 30 Hz window.

The probabilistic methodology for estimating the seismic hazard is adequate in this type of work, since rigorous consideration of the uncertainties associated with the different aspects of the occurrence of the movement is required. However, in future exploration missions to the Moon, it is necessary to carry out more studies of its internal structure and the implementation of a new generation of seismometers in a global seismic network. That will allow observation in a wider window of time, in order to characterize with greater certainty, the generation mechanisms of the Moonquakes.

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